

Romulo de Moraes

Software Developer

I am an Information Systems student with a strong interest in web development. On my GitHub, I stay active with personal and group projects with various goals and themes, using technologies similar to those mentioned in the experiences below. I am communicative, enjoy writing and teaching about what I learn, and I love challenges.

Address:

Rio Grande, Brazil

Email:

romulotg12@gmail.com

LinkedIn:

[LinkedIn Profile](#)

GitHub:

[GitHub Profile](#)

Portfolio:

romulodm.dev

Experiences



Byte Jr. Volunteer - Rio Grande, Brazil

President | March 2024 - Present (2 years 6 months)

People Management

Administration

I am working on restructuring the junior enterprise Byte Jr. together with Professor Dr. Luciano Maciel Ribeiro and colleagues from C3 - FURG. We are continuing projects that were on hold and improving the company's financial and administrative health.



iTEC - Plena Research Fellowship - Rio Grande, Brazil

Back-end Developer | April 2024 - Present (2 years 5 months)

NestJS

Jest

ESLint

Prisma

PostgreSQL

Azure Dev-Ops

Docker

Terraform

AWS

I am working on the backend development of an application focused on women's health. We are using microservices architecture with NestJS, working together with the computer vision, artificial intelligence, and frontend teams.



iTEC - MexTec Research Fellowship - Rio Grande, Brazil

Full-stack Developer | December 2023 - July 2024 (7 months)

React

Tailwind CSS

Material UI

React-Flow

Redux

Express

Mongoose

JWT

Socket.io

MQTT

ESP32

LoRa

I worked on developing a web system that processes and displays information on a control panel. These data were collected from various sensors and transmitted via ESP32. Additionally, I was involved in integration with the teams responsible for maintaining and operating the sensors.

- Development of a web system using React, with authentication, real-time data display, access levels...
- Backend development with Express, building some modules and adapting them for the frontend.
- Communication via ESP32 using HTTP and MQTT.

- Project infrastructure management, checking logs and performance.



EMAJ Volunteer - Rio Grande, Brazil

Full-stack Developer | August 2023 - December 2023 (4 months)

React Tailwind CSS Material UI Redux Express JWT Multer AWS S3 Sequelize
PostgreSQL

EMAJ is a legal advisory office at FURG run by law students and professors, serving the general public. I worked on developing a web system to assist EMAJ members in their daily activities. The idea came from Professor André Prisco, coordinator of the Information Systems course, and was part of the thesis of three Computer Engineering students.



LAMSA Volunteer - Rio Grande, Brazil

Unity Developer | May 2023 - November 2023 (6 months)

C# Unity

I assisted in researching and developing a virtual reality game using the Unity platform, addressing the theme of ocean preservation and aiming to educate children about the importance of conserving marine species.



PETC3 Research Fellowship - Rio Grande, Brazil

Researcher | June 2022 - December 2023 (1 year 6 months)

HTML CSS JavaScript Python Flask React Express JWT PostgreSQL MongoDB

PET is a government program aimed at groups of students who demonstrate potential, aiming to promote extracurricular activities under the guidance of a tutor professor. This project was my first experience within FURG and was essential for improving some soft skills - mainly communication and teamwork.

- Participation in the LEGO subproject, where we taught computational thinking to students in the region's schools using Scratch and LEGO Mindstorms.
- Participation in the PETCode subproject, where we created applications to assist PETC3's internal demands, posted on Instagram about programming, and studied together.
- Organization of events for the students of C3 - FURG.

Education



Universidade Federal do Rio Grande - FURG April 2026 - Present

Master's Degree | Computer Engineering (PPGComp)



Federal University of Rio Grande - FURG April 2022 - March 2026

Bachelor's Degree | Information Systems

Languages

Technical skills

Languages

Python, TypeScript/JavaScript, Go, C#, C++

Back-end

FastAPI, NestJS, Express, Flask, Prisma, Celery, Kafka, Redis

Data

PostgreSQL/PostGIS, MongoDB, H3, XGBoost/Random Forest

Front-end

React, Flutter, Tailwind CSS, Material UI

DevOps & Cloud

Docker, Kubernetes (KEDA), AWS (S3, SNS, ECS, IoT), Terraform, GitHub Actions, Sentry

IoT

MQTT, ESP32, Raspberry Pi, LoRa, Modbus, OPC-UA

Practices

Git, SCRUM, testing (Jest), Swagger/OpenAPI



Research interests

- Prize-linked savings systems
- Decentralized lottery mechanisms and verifiable on-chain randomness
- Geospatial processing and analysis of GNSS telemetry